EC241AT

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution Affiliated to VTU)

IV Semester B. E. Regular Examinations SEP/OCT – 2024

Electronics and Communication Engineering

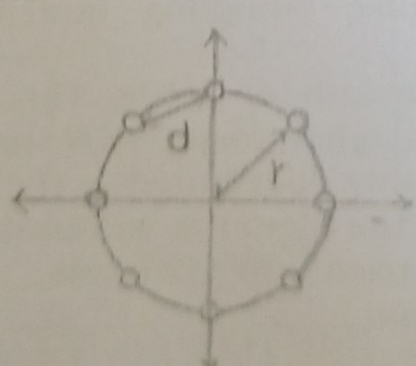
MATHEMATICS FOR COMMUNICATION ENGINEERING

Time: 03 Hours *Maximum Marks: 100*

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

PART-A

			M	BT	CO
1	1.1	The minimum step size required for a Delta – Modulator operating at $32K$sample/sec to track the signal $x(t) = 125 * t * [U(t) - U(t-1)] + (250 - 125 * t * [U(t) - U(t-2)]$ (Here $U(t)$ is the unit-step function so that slope overload is avoided) would be ________.	02	2	1
	1.2	Three analog signals, having bandwidths $1200Hz, 600Hz,$ are samples at their respective Nyquist rates, encoded with 12 bit words and then time-division multiplexed. Calculate the bit rate for the multiplexed signal.	02	2	1
	1.3	Let $X(t) = 8\cos(2\pi + \theta)$, where θ uniformly distributed in the interval $[0,2pi]$. The total power of the process = ________W.	02	2	2
	1.4	If the number of bits per sample in a Pulse Coded Modulation system is increased from $4\,bits$ to $8\,bits$, the improvement in signal to quantization noise ratio will be approximately ________ dB.	02	1	2
	1.5	A WSS random process $x(t)$ is applied to the input of an LTI system with impulse response $h(t) = 3e^{-2t} * U(t)$. Find the mean value of the output signal $Y(t)$ of the system if $E(x(t)) = 2$.	02	2	2
	1.6	Find the probability of error of $16\,PSK$ having symbol energy $-60\,dB$ and noise spectral density $\frac{N0}{2} = 0.5 * 10^{-6}\,w/Hz$.	02	2	3
	1.7	Consider the 8-phse constellation shown in fig 1.8, where d is the distance between two signaling points and r is the radius of the circle. Express r in terms of d.  Fig 1.7	02	2	3
	1.8	A binary PSK uses pulse of energy Eb and duration Tb. The symbols 1 and 0 are equiprobable and are statistically independent. The power spectral density of binary PSK is ________.	02	1	3
	1.9	A PAM signal with 8 levels has voltages $\{-14, -10, -6, -2, 2, 6, 10, 14\}$. Find the probability of error (optimistic estimation) if $N0/2 = 600 * 10^{-3} W/Hz$.	02	2	4

1.10 In the signals shown in fig 1.10 determine the sketch the basis functions.

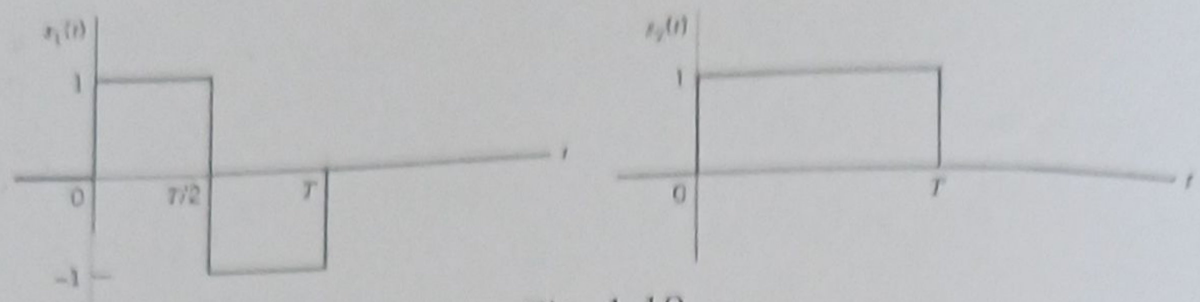


Fig. 1.10

(02 | 2 | 4)

PART-B

2 a Let $X(t)$ and $Y(t)$ be defined by

$X(t) = A\cos(wt) + Bsin(wt)$,

$Y(t) = B\cos(wt) - Asin(wt)$,

Where w is a constant and A and B are independent random variables both having variance σ^2. Find the cross-correlation $X(t)$ and $Y(t)$. (05 | 2 | 1)

b If $X(t) = A\cos(wt + \theta)$, where A and w are constants, θ is random variable uniformly distributed in the interval $[-\pi, \pi]$. Find if the random process $X(t)$ is ergodic with respect to mean and autocorrelation. (05 | 3 | 1)

c A stationary Gaussian random process $X(t)$ with zero mean and power spectral density $Sx(f)$ is applied to a linear filter whose impulse response is as shown in fig 2c. A sample Y is taken of the random process at the filter output at time T.

i) Determine the mean and variance of Y.
ii) What is the probability density function Y?

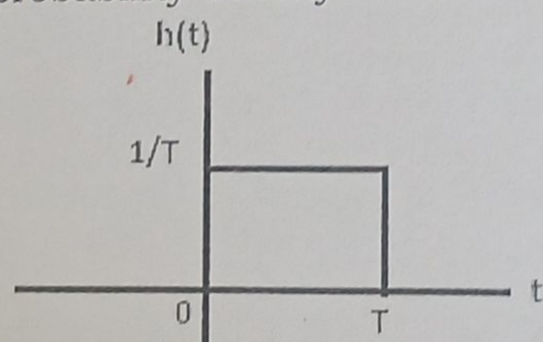


Fig. 2c

(06 | 3 | 1)

3 a State low pass sampling theorem. Deduce the expression for flat top sampling theorem in time and frequency domain. What is the difference between impulse sampling and flat top sampling? Illustrate with waveforms. (08 | 1 | 1)

b A video signal has bandwidth of $4MHz$. This signal is sampled, quantized and binary encoded using PCM technique. The video signal amplitude is uniformly distributed in the range $[-2,2]$ and an average power of $12dB$. If the signal is sampled at the Nyquist rate and each samples is quantized into 512 levels.

i) Determine the bit rate
ii) Determine the SNR
iii) Compute the minimum bandwidth required for the transmission
iv) If the SNR has to be increased by a factor of 16, how to achieve that? Will this affect the bit rate?

(08 | 3 | 1)

OR

Q. No.		Question	Marks	CO	BL
4	a	Deduce an expression for the Quantization noise of a robust quantizer. Check whether the performance of a uniform quantizer with companding is better than that of the uniform quantizer without companding.	08	1	2
	b	A delta modulator system is designed to operate at five times the Nyquist rate for a signal having a bandwidth equal to $3\,Khz$. Calculate the maximum amplitude of a $2\,Khz$ input sinusoidal signal for which the delta modulator does not have slope overload. Give that quantizing step size is $250\,mv$.	08	2	2
5	a	Five analog signals $S1$ and $S5$ in table 1 are to be formatted as digital signals for transmission. The allowed overhead includes an 8 bits unique word with $2\,bit$ frame sequence number. Design a multiplexing scheme clearly showing the framing and line rates. What should be the P/D buffer size at the receiver?	08	3	2

Table 1

Signals	*Bandwidth*	*Resolution lev*
$S1$	$36\,KHz$, sampled at 20% above Nyquist rate	1024
$S2\ \&\ S3$	$4\,KHz$, sampled at 5% above Nyquist rate	512
$S4$	$3\,KHz$, sampled at 10% above Nyquist rate	1024
$S5$	$5\,KHz$, sampled at 5% above Nyquist rate	4096

Q. No.		Question	Marks	CO	BL
	b	Express the real bandpass signal in terms of its complex lowpass equivalent. Deduce the I/Q modulator and demodulator architectures from the mathematical formulation.	08	1	2

OR

Q. No.		Question	Marks	CO	BL
6	a	With the help of Finite State Machine, describe the principal of operation of a P/D buffer implemented at receivers for quasi synchronous digital de-multiplexing scheme.	08	2	2
	b	Consider the signal $s1(t), s2(t), s3(t)$ *and* $s4(t)$ shown in fig 6b. Use gram Schmidt orthogonalization procedure to find an orthonormal basis for this set of signals. Express these signals in terms of basis functions and plot constellation diagram.	08	2	2

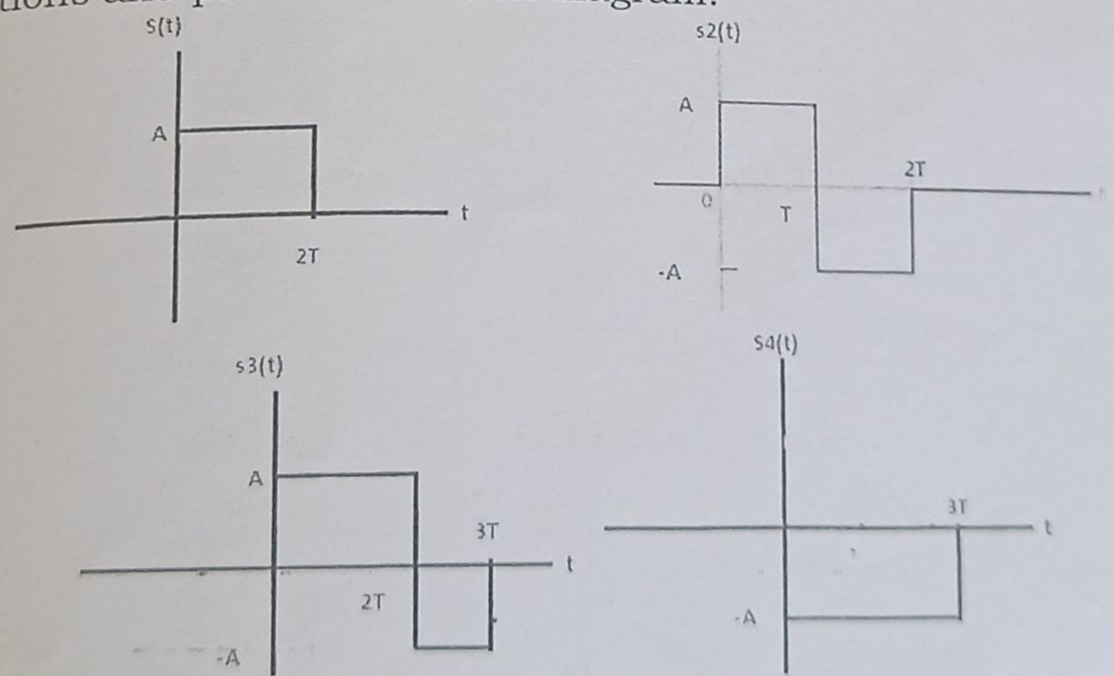
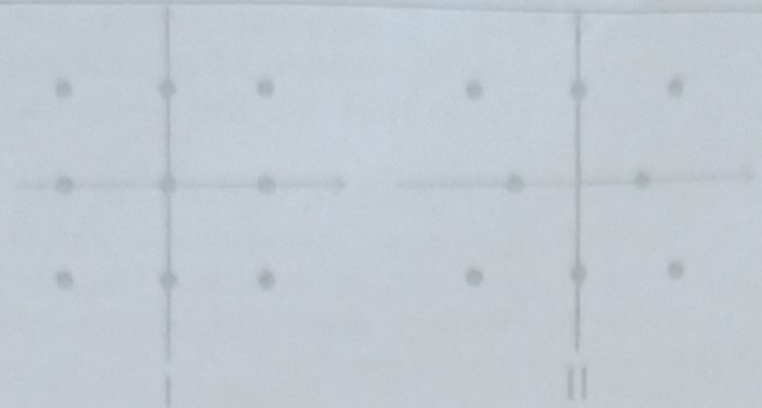
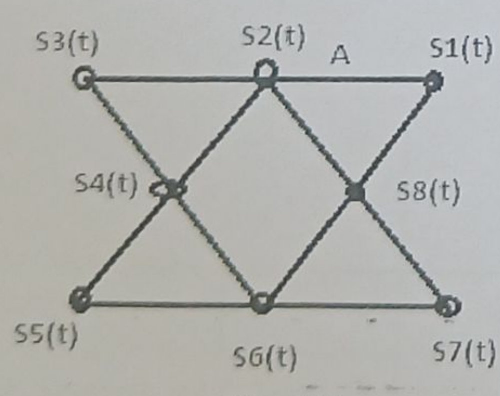
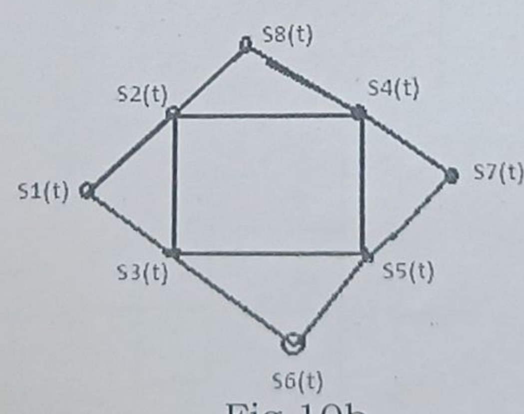


Fig. 6b

Q. No.		Question	Marks	CO	BL
7	a	Deduce the expression for the PSD of a Manchester Line code. Comment on the parameters (DC content, how the timing recovery at the receiver is managed).	08	2	3
	b	Consider the two 8-point QAM signal constellation shown in fig 7b. The minimum distance between adjacent points is 2A. Determine the average transmitted power for each constellation, assuming that the signal points are equally probable. Which constellation is more power-efficient?			

		 Fig. 7b	08	3	3
		OR			
8	a	Deduce the following for BFSK Modulation technique: i) Obtain the basis functions and express the signals in terms of basis functions. Also sketch the constellation. ii) Obtain the minimum Euclidian distance between the symbols and average symbol energy and draw transmitter architecture.	08	1	3
	b	Consider a 4 PSK constellation with points $(\sqrt{2},,j\sqrt{2},-\sqrt{2},-j\sqrt{2})$ and a 4PAM constellation $\{-3,-1,1,2\}$. If all the points in the constellation occur with equal probability. Find the ratio of the average energy of 4 PAM signal to that of a 4-PSK signal.	08	2	3
9	a	Deduce the probability of error expression in terms of the minimum distance between symbols. Obtain the probability of error expression for M-PSK and M-QAM.	08	2	4
	b	Consider two symbol cases with $P(s_1) = P(s_2)$. Obtain the expression for decision boundary (λ) for which the probability of error is minimum using clock sampling. Using the optimum λ, derive the Pe for two symbol case.	08	2	4
		OR			
10	a	For the 8-QAM signal constellation of fig 10a and fig 10b, the distance between any two adjacent points is 'A'. Obtain optimistic and pessimistic Pe for both the constellations.  Fig 10a  Fig 10b	08	3	4
	b	With the help of Block diagram discuss the optimum receiver architecture for M sybmols. Deduce the relevant mathematical expressions.	08	2	4